A One-Loop BV Obstruction in Pure Weyl Gravity and Its Formal τ-adic Wess–Zumino Resolution

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Working draft, July 2026

Abstract

We compute the coefficient-bearing one-loop BV anomaly of strict four- dimensional pure Weyl gravity in the complete gauge-fixed local quotient on the declared regular Bach-locus chart. The repository Euclidean elliptic complex, measure, zero-mode, contour, parity, and local heat-kernel ledgers give

$$\mathcal{A}^{(1)} = \frac{1}{(4\pi)^2}\left(\frac{199}{30}[\omega C^2] - \frac{87}{20}[\omega E_4]\right).$$

Both coordinates are nonzero in the strict quotient. Hence the fixed-field- content local Euclidean quantum master equation is obstructed at one loop. An exact dual-cone witness further excludes cancellation by any nonnegative combination of standard-sign free conformal scalars, Weyl or Dirac fermions, and gauge vectors.

We then adjoin a shifting scalar τ and derive its complete minimal-BV cotangent lift from $S_{\tau,\min} = \int \tau^*(\mathcal{L}_\xi \tau + \omega)$. In dressed canonical variables

$$\widehat{g} = e^{-2\tau} g, \qquad \widehat{g}^* = e^{2\tau} g^*, \qquad \widehat{\tau}^* = \tau^* + 2g_{\mu\nu} g^{*\mu\nu},$$

the Weyl sector is an exact contractible quartet. In the formal τ-adic local analytic jet algebra the complete dimension-four extended quotient is

$$\dim H^{0,4}_{\mathrm{ext}} = (3\text{ even}, 1\text{ odd}), \qquad H^{1,4}_{\mathrm{ext}} = 0.$$

The coefficient-bearing Wess–Zumino counterterm therefore restores the extended local Euclidean QME at one loop.

The two dispositions refer to different theories and are not contradictory: strict field content is obstructed, while the formal τ-adic compensator extension is locally restored at one loop. Neither statement is an all-loop or Lorentzian QME theorem.

AI-use disclosure and computational accountability

OpenAI GPT-5 was used substantively for mathematical synthesis, verification code, claim auditing, and manuscript drafting. The human author directed the programme and remains responsible for the claims, citations, and final text. All cohomological ranks and cancellations reported here use exact rational arithmetic. Every headline statement is bound to a deterministic certificate and an independent verifier. Submission remains conditional on final human author review.

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Contents

1 Question, result, and claim boundary

Pure Weyl gravity has classical Diff×Weyl gauge symmetry. The quantum question is not whether a determinant has a nonzero logarithm, but whether the regularized Slavnov breaking is trivial in the relative BV cohomology $H^{1,4}(s \mid \mathrm{d})$. We therefore separate three calculations:

1. classify the complete local obstruction space;
2. compute the coefficient vector in one specified analytic route;
3. decide whether that vector is a boundary in the chosen field theory.

The result has two theory labels which must not be suppressed:

Theory	One-loop local Euclidean disposition
strict pure Weyl gravity, fixed field content	obstructed
formal τ-adic Wess–Zumino compensator extension	QME restored

The second row does not reverse the first. It changes the BV complex.

All local results carry LOCAL-ALGEBRAIC; the coefficient computation also carries EUCLIDEAN-SPECTRAL. No result in this paper carries LORENTZIAN-CAUSAL.

2 Strict local BV obstruction space

The local jet algebra tracks ghost number, antifield number, form degree, engineering dimension, Grassmann parity, spacetime parity, and Weyl weight. Tensor symmetries, algebraic and differential Bianchi identities, graded commutativity, dummy-index relabelling, integration by parts, and total derivatives are reduced before exact sparse linear algebra is performed.

On the regular Bach-locus chart, the minimal Koszul–Tate sector contracts at positive antifield number. Ten nonminimal pairs contract explicitly, and an arbitrary invertible local BV-canonical gauge-fixing transformation preserves the quotient. The pure-diffeomorphism and mixed Diff–Weyl total complexes add no independent four-dimensional anomaly class.

Theorem 2.1 (Strict dimension-four anomaly quotient on the regular Bach locus)**.** *In the declared complete gauge-fixed local BV complex on a regular Bach-locus chart,*

$$\dim H^{1,4}_{\mathrm{even}}(s \mid \mathrm{d}) = 2, \qquad \dim H^{1,4}_{\mathrm{odd}}(s \mid \mathrm{d}) = 1.$$

Representatives may be chosen as

$$[\omega C^2], \qquad [\omega E_4], \qquad [\omega C\widetilde{C}].$$

The type-D carrier $\omega\Box R$ is exact with its stored R^2 counterterm and horizontal current.

The qualifier is load-bearing. In adapted jet coordinates the Bach Euler row and its differential consequences replace the transverse-tracefree highest metric jets. Rank-changing strata with additional reducibility or Killing data are not covered by this chart theorem.

The proof is an antifield-number spectral sequence rather than a list of familiar densities, following the local antifield-BRST architecture of Barnich, Brandt, and Henneaux [3]. Its exact steps are:

1. At E_0, the Koszul–Tate differential contracts the six adapted pairs (g^*, E_g), (ξ^*, N_ξ), (ω^*, N_ω) and their Lie-prolonged partners. Hence every positive-antifield column vanishes on the regular chart.

2. At E_1, the longitudinal differential reduces the antifield-zero sector to Diff$\times$Weyl invariant jets. The forward tensor-graph enumerator and reverse signature coverage agree under the dimension-four bound.

3. Relative d-cohomology and the total descent complex leave two even Weyl-ghost classes and one odd class. The pure-Diff and mixed total-complex blocks have zero quotient.

4. Ten pointwise nonminimal pairs contract, and the contraction transports through any invertible local BV-canonical gauge-fixing map. Therefore the minimal, nonminimal, and gauge-fixed quotients are chain-isomorphic.

The spectral sequence collapses at E_2. In the type-D row the stored primitive and horizontal current obey the local identity

$$\sqrt{g}\,\omega\Box R\,\mathrm{d}^4x = -\frac{1}{12}s\big(\sqrt{g}\,R^2\,\mathrm{d}^4x\big) - \mathrm{d}\star\big(R\,\mathrm{d}\omega - \omega\,\mathrm{d}R\big).$$

Its integral on a boundaryless carrier is therefore $\int\sqrt{g}\,\omega\Box R = -(1/12)s\int\sqrt{g}\,R^2$. The full ranks and pair counts appear in the generated spectral-sequence table of the computational supplement.

Closure, exact primitives, complete basis manifests, and normalized dual nonmembership witnesses are stored independently. The universal diffeomorphism descent and intrinsic Weyl descent are recorded separately; in particular, a nonzero Diff-horizontal tower is not called a nontrivial intrinsic Weyl descent.

This separation matches the algebraic Weyl-anomaly classification of Boulanger [4]: the Euler/type-A class is distinguished by a nontrivial intrinsic Weyl descent, whereas strictly Weyl-invariant type-B densities have trivial intrinsic Weyl descent. The present calculation has a different target and scope. It retains the universal Diff completion, the dynamical metric antifields, the Koszul–Tate filtration, and the explicit minimal-to-gauge-fixed BV comparison needed for the pure-gravity master equation.

3 Coefficient-bearing one-loop breaking

The coefficient route combines two independent curvature carriers. A local Ricci-flat carrier exposes C^2, while round S^4 fixes the Euler coordinate. The first five-dimensional bundle in the Euclidean symbol sequence is the gauge-parameter bundle $T \oplus \mathbf{1}$ (four Diff directions plus one Weyl direction), the ten-dimensional middle bundle is S^2T^*, and the final five-dimensional bundle is the transverse-tracefree symbol slice. Thus

$$(T \oplus \mathbf{1})_5 \xrightarrow{\sigma(K)} (S^2T^*)_{10} \xrightarrow{\sigma(P_{\mathrm{TT}})} (S^2_{\mathrm{TT}}T^*)_5, \qquad \operatorname{im}\sigma(K) = \ker\sigma(P_{\mathrm{TT}}).$$

The separate $5 \leftarrow 10 \leftarrow 5$ sequence is its cotangent formal adjoint; the endpoint is not being identified with the unreduced ten-component Bach equation. Determinant multiplicities, the functional measure, conformal-Killing zero modes, the factorwise spectral cut, the locally constant negative-scalar phase, and the parity Ward identity are included in the accepted integration slice.

The complete local determinant ledger is small enough to display:

sector	operator	order	rank	statistics	Z exponent	zero modes	local prescription
metric TT, upper	$\Delta_{2,\perp}(4)$	2	5	bosonic	$-1/2$	0	factorwise cut
Diff–Weyl scalar ghost	$\Delta_0(-4)'$	2	1	fermionic	$+1/2$	5	cut; negative-mode phase locally constant
metric TT, lower	$\Delta_{2,\perp}(2)$	2	5	bosonic	$-1/2$	0	factorwise cut
transverse Diff ghost	$\Delta_{1,\perp}(-3)'$	2	3	fermionic	$+1/2$	10	factorwise cut

All four factors use the same factorwise zeta/heat-kernel spectral cut. The single negative scalar phase is locally constant on the fixed stabilizer stratum and therefore drops out of the local b_4 variation; its global phase is not claimed. The primes remove precisely the fifteen conformal-Killing ghost modes. York/Hodge Jacobians and the ten nonminimal doublets have already been included before the four displayed factors are formed.

The same ledger carries the following exact local heat-kernel coordinates after determinant exponents, measure factors, and quartet cancellations:

sector	C^2	E_4	$C\widetilde{C}$	$\Box R$
metric TT, upper	37/24	$-31/72$	0	0
Diff–Weyl scalar ghost	89/120	$-269/360$	0	0
metric TT, lower	29/8	$-181/72$	0	0
transverse Diff ghost	29/40	$-79/120$	0	0
exact sum	199/30	$-87/20$	0	0

The first two columns are reconstructed factor by factor from two independent carriers: the Ricci-flat coefficient gives $c_i - a_i$, while the round-S^4 calculation gives a_i. Thus $c_i = (c_i - a_i) + a_i$ for every row, not merely after summation. The parity-odd zero is factorwise because every operator is a real parity-even tensor Laplacian. By contrast, the displayed $\Box R$ zero is a convention for the *total* local functional: the single allowed R^2 counterterm sets the summed type-D coordinate to zero. It is not four independently scheme-invariant factorwise $\Box R$ calculations. This is still a bosonic-Weyl-parameter Ward carrier, not yet a BV cohomology class. The next step performs the regulated Ward insertion and ghost replacement.

To make the determinant-to-BV bridge explicit, let σ denote a bosonic Weyl parameter. Dimensional regularization first gives the local supertrace

$$\Gamma^{(1)}_{\mathrm{div}}(d) = \frac{\mu^{d-4}}{(4\pi)^2(d-4)} \int \sqrt{g}\,[cC^2 - aE_4 + b\Box R + \text{BRST-exact rows}].$$

The regulated Ward insertion is not inferred from this scalar functional by name matching. It is computed as

$$\mathcal{S}_S \Gamma^{(1)}_{\mathrm{reg}} = \lim_{d\to 4} s_d \Gamma^{(1)}_{\mathrm{div}}(d),$$

where the $(d-4)$ Weyl variation of the dimensionally continued densities cancels the pole. Replacing σ by the Weyl ghost ω, adding the certified universal Diff completion, and reducing the exact, gauge-dependent, and total-derivative rows against the complete gauge-fixed $H^{1,4}(s \mid \mathrm{d})$ gives

$$\mathcal{S}_S \Gamma^{(1)}_{\mathrm{reg}} = \frac{1}{(4\pi)^2} \int \sqrt{g}\,\omega(cC^2 - aE_4 + b\Box R) + s\mathcal{B} + \mathrm{d}\mathcal{C}.$$

More explicitly, for the continued integrated carriers $I_i(d)$ used by the regulator, the cohomological residue identity has the form

$$s_d I_i(d) = (d-4)A^4_i + s_d B_i(d) + \mathrm{d}C_i(d) + O\big((d-4)^2\big),$$

where A^4_i is the Diff-completed ghost insertion represented at top form by ωC^2, ωE_4, or $\omega\Box R$. Multiplication by the pole and passage to the quotient therefore sends each heat-kernel coordinate to the corresponding normalized dual coordinate in $H^{1,4}(s \mid \mathrm{d})$. This residue map, rather than equality of familiar coefficient names, is the determinant-to-Slavnov bridge. This is the coefficient-bearing map from the Euclidean integration slice to a ghost-number-one Slavnov class; it is why a logarithmic divergence, trace anomaly, and BV anomaly are not silently conflated. The complete reduction can be summarized by the sequence

$$\begin{tikzcd}
\Gamma^{(1)}_{\mathrm{div}}(d) \arrow[r, "s_d \text{ and } \mathrm{Res}_{d=4}"] & \mathcal{W}^{(1)}_\sigma \arrow[d, "\sigma\mapsto\omega + \text{Diff completion}"] & \\
& Z^{1,4}(s \mid \mathrm{d}) \arrow[r] & H^{1,4}(s \mid \mathrm{d}).
\end{tikzcd}$$

Here $\mathcal{W}^{(1)}_\sigma$ is the bosonic Ward insertion, the lower-left object is a ghost-number-one BV cocycle, and only its image under the final quotient arrow is the anomaly class. That last arrow removes explicit Slavnov-exact and horizontal rows. No determinant factor or heat-kernel coordinate is itself promoted directly to the class.

Theorem 3.1 (Repository one-loop Slavnov breaking)**.** *In the normalization*

$$S_W = \alpha_C \int \sqrt{g}\,C^2,$$

the repository-regulated breaking has coordinates

$$(C^2, E_4, C\widetilde{C}, \Box R) = \left(\frac{199}{30}, -\frac{87}{20}, 0, 0\right).$$

After reduction against the complete strict quotient,

$$\mathcal{A}^{(1)} = \frac{1}{(4\pi)^2}\left(\frac{199}{30}[\omega C^2] - \frac{87}{20}[\omega E_4]\right).$$

Consequently the strict fixed-field-content local Euclidean QME is obstructed at one loop.

The odd zero is derived from the parity Ward identity for the declared real tensor-Laplacian regulator. The type-D zero is scheme-dependent but exact; it cannot cancel either nontrivial coordinate. A beta function alone would not prove this theorem: the decisive object is the regulated Slavnov breaking reduced in the complete BV quotient.

The even numbers agree, after translating the Euler-sign convention, with the classic one-loop Weyl-gravity anomaly of Fradkin and Tseytlin [5]. Our claim is not that these constants were previously unknown. It is the content-addressed identification of the repository elliptic complex, multiplicities, measure, zero modes, contour, and regulator with a complete BV obstruction basis, followed by the explicit Slavnov-class disposition. The local $\Box R$ term has a separate scheme-sensitive history; the recent heat-kernel analysis of Barvinsky et al. [7] makes that distinction explicit. In the scheme frozen here its coefficient is zero and the class is in any case removed by the stored R^2 primitive.

Proposition 3.2 (Standard unitary free matter cannot cancel the vector)**.** *Let the added matter content be any nonnegative combination of real conformal scalars, Weyl or Dirac fermions, and standard gauge vectors. No such combination cancels both even coordinates of the strict breaking.*

These are free, massless spectator fields conformally coupled to the same metric; gauge vectors include their standard BRST ghosts. In the convention $(c, -a)$ their exact vectors are

field	c	$-a$
pure Weyl graviton	199/30	−87/20
real conformal scalar	1/120	−1/360
Weyl fermion	1/40	−11/720
Dirac fermion	1/20	−11/360
gauge vector	1/10	−31/180

The separating functional is simply $\ell(c, -a) = c$: it is strictly positive on the gravity vector and on every allowed matter ray. Hence no nonnegative real combination, and a fortiori no nonnegative integer field content, can reach the origin. Interacting fixed points, wrong-sign or nonunitary matter, conformal higher spins, and compensator extensions lie outside this no-go.

4 The compensator cotangent lift

Adjoin a scalar with BRST transformation

$$Q\tau = \mathcal{L}_\xi \tau + \omega.$$

Its cotangent rows are not guessed. They are the Euler derivatives of the single minimal master term

$$S_{\tau,\mathrm{min}} = \int \tau^*(\mathcal{L}_\xi \tau + \omega).$$

In addition to the field row this forces

$$\delta\omega^* = 2g_{\mu\nu}g^{*\mu\nu} + \tau^*, \qquad \delta\xi^*_\mu = N_{\xi,\mu} + \tau^*\nabla_\mu\tau, \qquad \gamma\tau^* = \mathcal{L}_\xi\tau^*,$$

together with the opposite-adjoint Lie-prolonged rows.

Theorem 4.1 (Exact minimal-BV cotangent lift). *On every strict-plus-compensator generator and covariant jet generator,*

$$\delta^2 = 0, \qquad \delta\gamma + \gamma\delta = 0, \qquad Q^2 = 0.$$

The canonical change

$$\widehat{g} = e^{-2\tau} g, \quad \widehat{g}^* = e^{2\tau} g^*, \quad \widehat{\tau}^* = \tau^* + 2g_{\mu\nu} g^{*\mu\nu}$$

preserves the canonical one-form and sends the Weyl Koszul–Tate row to $\delta\omega^* = \widehat{\tau}^*$. Thus $(\tau, \omega, \omega^*, \widehat{\tau}^*)$ is an exact contractible quartet.

In the ordered basis $(\tau, \omega, \omega^*, \widehat{\tau}^*)$, the Weyl part of the differential and its homotopy act by

$$Q_W \tau = \omega, \qquad Q_W \omega^* = \widehat{\tau}^*, \qquad h\omega = \tau, \qquad h\widehat{\tau}^* = \omega^*,$$

and vanish on the other two basis generators. Consequently

$$Q_W h + h Q_W = N_{\text{quartet}}.$$

On a monomial of positive quartet number the contracting homotopy is $N_{\text{quartet}}^{-1} h$, extended to covariant jets as a graded derivation. The Diff and horizontal pieces preserve quartet number, so the filtered total-complex contraction follows by homological perturbation in the declared formal completion.

For clarity, the total-complex contraction can be read without replaying the boundary matrix. Write $\mathcal{D} = Q_W + \delta$, with δ the remaining Diff, Koszul–Tate, and horizontal rows in dressed variables. The four quartet members occur in matched covariant Diff modules, and h is extended to their jets as a covariant graded derivation. Hence $[N_{\text{quartet}}, \delta] = 0$ and the graded anticommutator $[\delta, h]_+$ vanishes. Therefore

$$\mathcal{D}h + h\mathcal{D} = N_{\text{quartet}}.$$

If $\mathcal{D}\mathbf{a} = 0$ and $\mathbf{a}_n$ has positive quartet number n, then $\mathbf{a}_n = \mathcal{D}(n^{-1} h \mathbf{a}_n)$. The total cohomology is consequently represented at quartet number zero, where $\mathcal{D}$ is the pure-Diff BV total differential of $\widehat{g}$. Thus $H^{1,4}_{\text{ext}} = 0$ imports the independently certified pure-Diff ghost-one result; it is not inferred only from the four displayed Wess–Zumino primitives.

The exponential change is an automorphism only after declaring the formal τ-adic local analytic completion. Its two inverse series are replayed with exact rational coefficients and obey at every order

$$\sum_{k=0}^{n} \frac{(-2)^k}{k!} \frac{2^{n-k}}{(n-k)!} = \frac{2^n}{n!}(1-1)^n = \delta_{n0}.$$

No finite-polynomial-in-τ theorem is inferred.

5 Extended cohomology and QME restoration

Quartet contraction reduces the extended complex to pure Diff BV cohomology in the dressed metric $\widehat{g}$. The dimension-four counterterm inventory is therefore larger than in strict Weyl gravity: $R(\widehat{g})^2$ is now closed. The declared algebra is the formal τ-adic local analytic jet completion generated by $\widehat{g}$, its cotangent variables, Diff ghosts, the quartet, curvatures and covariant derivatives, with the same dimension and form-degree bounds as the strict calculation. The canonical dressed change is invertible in this completion. After quartet contraction, forward invariant generation and reverse grading-signature coverage leave precisely the pure-Diff dimension-four metric invariants below; the independently completed pure-Diff ghost-one total complex has zero quotient. Nonminimal and gauge-fixing transport then preserve that result. Thus the vanishing assertion is a complete statement in this declared formal algebra, not an AFN0 primitive test alone.

Theorem 5.1 (Formal τ-adic extended quotient)**.** *In the declared formal τ-adic local analytic jet algebra on the regular Bach locus,*

$$H^{0,4}_{\text{ext,even}} = \text{span}\{C(\widehat{g})^2, E_4(\widehat{g}), R(\widehat{g})^2\},$$

$$H^{0,4}_{\text{ext,odd}} = \text{span}\{C(\widehat{g})\widetilde{C}(\widehat{g})\}, \qquad H^{1,4}_{\text{ext}} = 0.$$

$\Box R(\widehat{g})$ remains horizontally exact. There is no incoming dimension-four positive-antifield boundary: the only dimension-zero natural symmetric tensor is $\widehat{g}$, whose image is the vanishing trace of the Bach row.

Explicit primitives for the strict four-candidate anomaly carrier are

$$B_C = \int \sqrt{g}\,\tau C^2, \qquad B_P = \int \sqrt{g}\,\tau C\widetilde{C},$$

and the Euler Wess–Zumino functional is

$$B_E = \int \sqrt{g}\,\left[\tau E_4 + 4G^{\mu\nu}\nabla_\mu\tau\nabla_\nu\tau - 4(\Box\tau)(\nabla\tau)^2 + 2(\nabla\tau)^4\right].$$

With the conventions above,

$$s_{\text{ext}}B_E = \int \sqrt{g}\,\omega E_4 \pmod{\mathrm{d}}.$$

The termwise intrinsic Euler descent and its universal Diff completion are stored separately. Together with the R^2 primitive $B_\Box$, these four primitives have an exact rank-four boundary matrix. The curved-background Euler primitive agrees with the standard dilaton Wess–Zumino construction obtained from finite Weyl transformation of dimensionally regulated counterterms [6]. The new certificate needed here is not the displayed AFN0 formula alone: it is the cotangent lift, quartet contraction, exhaustive extended $H^{0,4}/H^{1,4}$ comparison, and coefficient-bearing BV boundary equation.

This restoration has a physical price. The compensator theory is not quantum-equivalent to strict Weyl gravity: gauge fixing $\tau = 0$ does not turn the anomalous strict quotient into an anomaly-free theory, while the dressed Diff theory permits counterterms such as $R(\widehat{g})^2$ and finite normalizations unavailable in the strict quotient. Exactness is obtained by enlarging both the BV complex and its local counterterm algebra.

Corollary 5.2 (One-loop local Euclidean QME restoration)**.** *The counterterm*

$$-\frac{\hbar}{(4\pi)^2}\left(\frac{199}{30}B_C - \frac{87}{20}B_E\right)$$

satisfies

$$\mathcal{A}^{(1)} + s_{\text{ext}}S^{(1)}_{\text{ct}} = 0 \pmod{\mathrm{d}}.$$

Hence the compensator-extended local Euclidean QME is restored at one loop in the declared formal τ-adic algebra.

Proposition 5.3 (The restored QME does not determine a unique Q_1)**.** *Define the coefficient-bearing one-loop Wess–Zumino functional by*

$$C_1^{\text{WZ}} = -\frac{1}{(4\pi)^2}\left(\frac{199}{30}B_C - \frac{87}{20}B_E\right).$$

The local Wess–Zumino Hamiltonian contribution

$$Q_1^{\mathrm{WZ}} = (C_1^{\mathrm{WZ}}, \cdot)$$

is coefficient-bearing and fixed by the counterterm above. The complete renormalized first Slavnov operator is nevertheless underdetermined. On the dimensionless flat Euclidean fixture $p = (1, 0, 0, 0)$*, in units of the chosen reference momentum, with* $h_{11} = 1$*,* $h_{22} = -1$ *and all other TT components zero, together with one conformal polarization, the bulk quadratic response of the allowed basis* $(C(\widehat{g})^2, E_4(\widehat{g}), R(\widehat{g})^2, C(\widehat{g})\widetilde{C}(\widehat{g}))$ *is*

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 9 & 0 \end{pmatrix}.$$

Thus the C^2 *and* R^2 *finite-counterterm directions change the bulk Hamiltonian vector field independently. This response calculation alone does not supply a nonlocal representative, the renormalized BV Laplacian or time-ordered product, or finite normalization conditions. The next proposition supplies the anomaly-induced nonlocal component, but not the Weyl-invariant finite remainder.*

6 Consequences for one-loop effective-action representatives

Proposition 6.1 (Anomaly-induced Euclidean representative)**.** *Set*

$$\mathcal{E}_4 = E_4 - \frac{2}{3}\Box R, \qquad \Delta_4 = \Box^2 + 2R^{\mu\nu}\nabla_\mu\nabla_\nu - \frac{2}{3}R\Box + \frac{1}{3}(\nabla^\mu R)\nabla_\mu.$$

This is the four-dimensional Paneitz operator [9]. On an invertible Euclidean boundary problem, or on the compatible source sector of a self-adjoint generalized inverse G_4 *of* Δ_4*, one anomaly-induced representative is*

$$\Gamma_{1,\mathrm{anom}} = \frac{1}{(4\pi)^2}\left\{\frac{1}{8}\left\langle \mathcal{E}_4, G_4\left(2cC^2 - a\mathcal{E}_4\right)\right\rangle + \frac{a}{18}\int \sqrt{g}\, R^2\right\}, \qquad (c, a) = \left(\frac{199}{30}, \frac{87}{20}\right).$$

Its Weyl variation is the certified anomaly $(4\pi)^{-2}(cC^2 - aE_4)$ *in that sector. The three functional coefficients are*

$$\left(\frac{199}{120}, -\frac{87}{160}, \frac{29}{120}\right).$$

The last coefficient follows from $\delta_\sigma \int \sqrt{g}R^2 = -12\int \sqrt{g}\,\sigma\Box R$ *and is required to return from the canonical* $\mathcal{E}_4$ *representative to the repository* $\Box R = 0$ *convention. This is the Riegert construction [8] with the scheme conversion made explicit; it is not the complete finite effective action.*

Proposition 6.2 (Universal flat-TT logarithmic form factor)**.** *On the positive nonzero-momentum flat Euclidean transverse-traceless carrier, the quadratic conformal-sector form factor is*

$$\Gamma^{(2)}_{1,\mathrm{TT}} = \frac{1}{(4\pi)^2}\left\langle C^{(1)}, F_C(p^2;\mu)C^{(1)}\right\rangle, \qquad F_C(p^2;\mu) = -\frac{199}{60}\log\frac{p^2}{\mu^2} + z_C(\mu).$$

The logarithmic coefficient obeys

$$-\frac{199}{60} = -\frac{c}{2} = -\frac{\beta_2}{4}, \qquad (c, \beta_2) = \left(\frac{199}{30}, \frac{199}{15}\right),$$

and therefore

$$\mu\partial_\mu F_C = \frac{199}{30}, \qquad F_C(p^2;\mu) - F_C(q^2;\mu) = -\frac{199}{60}\log\frac{p^2}{q^2}.$$

The factor of $1/2$ is fixed by the stated normalization: $\langle C^{(1)}, C^{(1)}\rangle = 1$ and $\mu\partial_\mu \log(p^2/\mu^2) = -2$, so matching the scale response c requires $-2A_{\log} = c$ and hence $A_{\log} = -c/2 = -199/60$. This fixes the universal scale-dependent part and the scheme-independent momentum difference. A reference condition $F_C(\kappa^2_{\rm ref}; \mu) = N_C$ instead fixes the still-free local constant as

$$z_C(\mu) = N_C + \frac{199}{60} \log \frac{\kappa^2_{\rm ref}}{\mu^2}.$$

The separation of anomaly-induced and conformal parts and the appearance of logarithmic non-local curvature form factors follow the covariant effective- action framework of Barvinsky and Wachowski [10]. The present certificate is restricted to the declared flat carrier. The next two propositions supply its curvature-squared covariantization and one exact Fradkin–Vilkovisky completion of that selected carrier. They do not compute the third-curvature repository form-factor functions or coefficients, a zero-momentum definition, Lorentzian branch prescription, or a particle state.

Proposition 6.3 (Endomorphism-class covariant logarithm through curvature order two)**.** *Let Δ_C be a positive self-adjoint Laplace-type operator on the algebraic Weyl-tensor bundle, with a fixed self-adjoint elliptic domain, and let $\Pi_\perp$ project off its kernel. Covariant perturbation theory through total curvature order two gives*

$$\Gamma^{(2)}_{1, C^2 \log} = -\frac{1}{(4\pi)^2} \frac{199}{60} \left\langle C, \Pi_\perp \log \frac{\Delta_C}{\mu^2} \Pi_\perp C \right\rangle.$$

It obeys

$$\mu\partial_\mu \log(\Delta_C/\mu^2) = -2\Pi_\perp,$$

and hence has the certified scale response $(4\pi)^{-2}(199/30)\langle C, \Pi_\perp C\rangle$.

This quadratic-curvature functional is independent, at the declared order, within the class of operators sharing the same principal symbol, connection, self-adjoint domain, and source-complement prescription and differing at first curvature order only by a bundle endomorphism. Indeed, if $\Delta'_C = \Delta_C + V_1 + O(\mathcal{R}^2)$ with V_1 a self-adjoint curvature-order-one bundle endomorphism on the same domain, then

$$d \log(\Delta_C)[V_1] = \int_0^\infty (\Delta_C + s)^{-1} V_1 (\Delta_C + s)^{-1} \, ds.$$

Sandwiching this variation between two Weyl tensors has curvature order $1 + 1 + 1 = 3$. Thus the first missing completion data for this C^2 logarithmic carrier are cubic in the curvature [11, 14]. No independence under a change of bundle connection is claimed; such a change can introduce first-order differential terms not covered by V_1. Finite C^2 and absolute dressed $R(\widehat{g})^2$ normalizations, third-curvature repository functions and coefficients, and global kernel data remain open. The following proposition supplies a Weyl-orbit completion only for the selected C^2 logarithmic carrier.

Proposition 6.4 (Fradkin–Vilkovisky completion of the selected logarithmic carrier)**.** *Suppose the Euclidean conformal scalar operator*

$$L_g = \Box_g - \frac{1}{6} R_g$$

has a normalized inverse on an asymptotically flat or fixed-boundary function space, with no normalized zero mode. Define

$$u_g = 1 + \frac{1}{6} L_g^{-1} R_g, \qquad \Sigma_{\rm FV}[g] = -\log u_g, \qquad \bar{g}[g] = u_g^2 g.$$

Then $L_g u_g = 0$ *and*

$$R[\bar{g}] = u_g^{-3}(R_g - 6\Box_g)u_g = 0.$$

For boundary-compatible $g' = e^{2\sigma}g$*, conformal covariance and the normalization imply* $u_{g'} = e^{-\sigma}u_g$*, hence* $\bar{g}[g'] = \bar{g}[g]$*. Therefore*

$$\Gamma^{\rm conf}_{1,C^2}[g] = -\frac{1}{(4\pi)^2}\frac{199}{60}\left\langle C[\bar{g}], \Pi_{\bar{g}} \log\frac{\Delta_C[\bar{g}]}{\mu^2}\Pi_{\bar{g}}C[\bar{g}]\right\rangle_{\bar{g}}$$

is exactly invariant under the declared local Weyl transformations. Every tensor, pairing, projector, operator domain and spectral logarithm is evaluated on the same representative. This is a nonlocal functional invariant under local Weyl transformations, not a local functional.

Since $\Sigma_{\rm FV} = -(1/6)\Box^{-1}R + O(\mathcal{R}^2)$*, this completion agrees with the preceding carrier through curvature order two and its first forced correction is cubic. The Fréchet derivative of the spectral logarithm identifies that correction carrier, including transported pairing and projector terms, but does not determine the derivative-decorated nonlocal cubic Weyl-invariant form factors of the full one-loop action [10]. If* L_g *has a kernel, an orbit-compatible projector and normalization are additional data and the proposition does not apply as stated.*

Finally, $\bar{g}[g]$ *is not the compensator-dressed* $\widehat{g}[g,\tau] = e^{-2\tau}g$*: the former is a nonlocal functional of the strict metric fixed by scalar-flat gauge, while* τ *is an independent local field in the enlarged formal BV theory.*

Proposition 6.5 (FV anomaly action and dependent Ricci sector)**.** *Write* $\mathcal{E}_4 = E_4 - \frac{2}{3}\Box R$ *and use the same normalized FV orbit coordinate as above. The exact coefficient-bearing anomaly action is*

$$\Gamma_{1,\rm FV} = \frac{1}{(4\pi)^2}\int\sqrt{g}\left[\left(\frac{199}{30}C^2 - \frac{87}{20}\mathcal{E}_4\right)\Sigma_{\rm FV} + \frac{87}{10}\Sigma_{\rm FV}\Delta_4\Sigma_{\rm FV} + \frac{29}{120}R^2\right].$$

Under $\delta_\sigma\Sigma_{\rm FV} = \sigma$*, its non-target response cancels coefficientwise:*

$$-\frac{87}{5} + \frac{87}{5} = 0, \qquad \frac{29}{10} - \frac{29}{10} = 0.$$

Hence

$$\delta_\sigma\Gamma_{1,\rm FV} = \frac{1}{(4\pi)^2}\int\sqrt{g}\,\sigma\left(\frac{199}{30}C^2 - \frac{87}{20}E_4\right).$$

The specialization $\Sigma_{\rm RFT} = \frac{1}{4}G_4\mathcal{E}_4$ *reproduces exactly* $(199/120, -87/160, 29/120)$ *in the earlier Paneitz/Riegert basis.*

Moreover, in the declared massless conformal FV decomposition

$$\Gamma_1[g] = \Gamma_{1,\rm FV}[g] + W_1[\bar{g}[g]], \qquad R[\bar{g}] = 0,$$

the conformal functional W_1 *is generated by Weyl-curvature invariants before evaluation on* $\bar{g}$*. Therefore terms containing a Ricci scalar in a generic original-metric basis are determined by* $\Gamma_{1,\rm FV}$ *and the re-expansion of* $W_1[\bar{g}]$*. A term displayed as* $RF(\Box)R$ *may occur in that basis, but its form factor is not a separately specifiable datum in this decomposition [10]. This statement is restricted to the normalized asymptotically-flat or fixed-boundary FV domain and does not extend without new analysis to massive, explicitly nonconformal, or symmetry-broken theories.*

Proposition 6.6 (Algebraic cubic Weyl carriers in four dimensions)**.** *In the pointwise zero-derivative sector of three algebraic Weyl tensors in four Euclidean dimensions, the parity-even and parity-odd carrier spaces are both one-dimensional:*

$$I_e = C_{ab}{}^{cd}C_{cd}{}^{ef}C_{ef}{}^{ab}, \qquad I_o = (*C)_{ab}{}^{cd}C_{cd}{}^{ef}C_{ef}{}^{ab}.$$

Equivalently, after the Hodge decomposition on two-forms, the chiral basis is $I_+ = \operatorname{tr}(C_+^3)$ *and* $I_- = \operatorname{tr}(C_-^3)$*, with*

$$\begin{pmatrix} I_+ \\ I_- \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} I_e \\ I_o \end{pmatrix}.$$

The exact contraction enumeration has fifteen raw six-slot pairings, three identical-factor orbits, two tracefree-zero orbits, and one surviving triangle $\operatorname{tr}(C_\pm^3)$ *in each chirality. The mixed allocations vanish because the chiral projectors are orthogonal and the resulting lone block can only close through its zero trace. An exact rank-two evaluation witness and the independent Schouten-zero Weyl quotient verify the parity basis.*

This is a LOCAL-ALGEBRAIC *carrier classification, not a third-order effective-action calculation. Since* C^3 *has engineering dimension six, it is not an additional dimension-four one-loop local counterterm. The derivative-decorated nonlocal carriers, functions* $\Gamma_m(\Box_1, \Box_2, \Box_3)$*, branch and permutation data, and all cubic coefficients remain open. In particular, unrestricted local integration by parts cannot be used across independently labelled* $\Box_i$ *[11, 10].*

Proposition 6.7 (Parity-even third-curvature carrier manifest)**.** *On the declared asymptotically flat scalar-flat Euclidean conformal representative, set*

$$K_{\mu\nu} = \frac{2}{\Box}\nabla^\beta \nabla^\alpha C_{\alpha\mu\beta\nu}.$$

The parity-even pure-gravity third-curvature sector has five carrier-labelled functions, denoted $I_{10}, I_{24}, I_{25}, I_{28}, I_{29}$*, in the covariant-perturbation-theory basis. Their schematic tensor rows and label stabilizers are*

I_{10}	$\operatorname{tr}(K_1K_2K_3)$	S_3
I_{24}	$K_1\nabla K_2\nabla K_3$	$S_2(23)$
I_{25}	$K_1\nabla K_2\nabla K_3$ *(crossed)*	$S_2(23)$
I_{28}	$\nabla K_1\nabla K_2\nabla\nabla K_3$	$S_2(12)$
I_{29}	$\nabla\nabla K_1\nabla\nabla K_2\nabla\nabla K_3$	C_3 (source), S_3 (scalar– flat).

The last enhancement is not assumed from the source notation. In Fourier space the transverse scalar-flat carrier factorizes as

$$I_{29}(1,2,3) = -(k_2K_1k_2)(k_3K_2k_3)(k_1K_3k_1).$$

Momentum conservation and $k_iK_i = 0$ *identify it with* $I_{29}(1,3,2)$*. Thus the effective scalar-flat label modules have dimensions* $1, 3, 3, 3, 1$*, with total dimension eleven and* S_3*-character* $(11, 5, 2)$*, or* $5\mathbf{1} \oplus 3\mathbf{2}_{\text{std}}$*. The source-generic* C_3 *stabilizer remains recorded separately.*

In four dimensions there is one integrated identity for every completely symmetric $F(\Box_1, \Box_2, \Box_3)$*. Writing* $x_i = \Box_i$*, its pure-*K *part is proportional to*

$$-\frac{1}{12}\big(x_1^2 + x_2^2 + x_3^2 - 2x_1x_2 - 2x_2x_3 - 2x_1x_3\big)I_{10} - \frac{1}{2}x_1 I_{24} - \frac{1}{2}(x_2 + x_3 - x_1)I_{25} + I_{28},$$

modulo total derivatives and $O(\mathcal{R}^4)$*;* I_{29} *is absent. Removing the fully symmetric component of* I_{28} *therefore leaves a ten-dimensional effective scalar-flat label module with character* $(10, 4, 1)$*, or*

$$4\mathbf{1} \oplus 3\mathbf{2}_{\text{std}}.$$

The I_{29} *row is the ineliminable lineage of the local parity-even algebraic* C^3 *carrier, but its normalization in the two conventions is not inferred [11, 13].*

This is a EUCLIDEAN-SPECTRAL *carrier and permutation- quotient theorem. It is five carrier-labelled functions, with the effective I_{29} symmetry and one functional relation, not ten computed form factors. The five repository functions, all their coefficients, and the parity-odd derivative-decorated carrier manifest remain open.*

Proposition 6.8 (Universal CPT kernels and the repository matching boundary)**.** *For the rank-one minimal scalar-Laplacian fixture $F = -\Box, P = 0, \mathcal{R}_{\mu\nu} = 0$, the five source functions associated with $I_{10}, I_{24}, I_{25}, I_{28}, I_{29}$ are imported exactly in the alpha-parameter convention*

$$\Gamma_i = \left\langle \frac{df\, f_i}{-\Omega} \right\rangle_3 + T_i + L_i, \qquad \Omega = \alpha_2\alpha_3\Box_1 + \alpha_1\alpha_3\Box_2 + \alpha_1\alpha_2\Box_3.$$

Their stabilizers and box homogeneities are

i	10	24	25	28	29
$\mathrm{Stab}_{\mathrm{source}}(I_i)$	S_3	$S_2(23)$	$S_2(23)$	$S_2(12)$	C_3
$\mathrm{Stab}_{K,\mathrm{sf}}(I_i)$	S_3	$S_2(23)$	$S_2(23)$	$S_2(12)$	S_3
$\deg_\Box \Gamma_i$	-1	-2	-2	-3	$-4.$

The exact source formulas, symmetrized rows and formula hashes are retained in the computational receipt [11].

This imports universal kernels, not the pure-Weyl conformal-graviton functions. For tensor and ghost Laplace-type blocks, both the endomorphism P and bundle curvature $\mathcal{R}_{\mu\nu}$ are already linear in the spacetime curvature and feed the same pure-gravity carriers through other CPT rows. The available round-sphere constrained determinant factors, Ricci-flat local b_4 match and principal-symbol sequence do not determine those generic-background trace substitutions. Consequently no repository third-curvature function or coefficient is inferred by summing bundle ranks. The minimal missing physical input is a same-gauge generic-background full-BV Hessian reduced to minimal Laplace-type blocks with all trace substitutions through curvature order three, or an equivalent direct nonminimal fourth-order CPT kernel, together with the matching measure.

Proposition 6.9 (Generic ghost obstruction to direct minimal-CPT matching)**.** *For the covariant gauge rows*

$$\delta F_\mu = \Box\xi_\mu + R_{\mu\nu}\xi^\nu + (1-2\beta)\nabla_\mu\nabla\cdot\xi + 2(1-4\beta)\nabla_\mu\omega, \qquad \delta F_W = 2\nabla\cdot\xi + 8\omega,$$

the algebraic Weyl-ghost Schur complement is independent of β:

$$(M_{\mathrm{eff}}\xi)_\mu = \Box\xi_\mu + R_{\mu\nu}\xi^\nu + \frac{1}{2}\nabla_\mu\nabla\cdot\xi.$$

At a unit covector its principal eigenvalues are $(3/2, 1, 1, 1)$, so the operator is elliptic but not of Laplace type. No fixed invertible left and right bundle maps can scalarize this symbol for all covectors: for two orthogonal unit covectors the exact comparison spectrum is $(2/3, 3/2, 1, 1)$. Moreover,

$$M_{\mathrm{eff}}(\nabla c) = -\frac{3}{2}\nabla\Delta_0 c + 2R_{\mu\nu}\nabla^\nu c,$$

so generic tracefree Ricci curvature sends a longitudinal input into a transverse component. The Hodge determinant split used by the Einstein four-factor ledger therefore does not extend covariantly. On an Einstein background it reduces exactly to

$$M_{\mathrm{eff}}(\nabla c) = -\frac{3}{2}\nabla(\Delta_0 - R/3)c,$$

recovering the accepted scalar ghost factor.

Consequently the imported minimal-Laplace CPT kernels cannot be applied to the generic repository ghost sector by direct rank or endomorphism substitution. The missing coefficient-bearing input is a matched nonminimal-vector determinant/CPT calculation, or an exact local extension with its determinant and Jacobian proved equivalent, together with the generic physical fourth-order Hessian kernel. This is an obstruction to the current matching architecture, not a new anomaly, QME, or Lorentzian no-go theorem.

Proposition 6.10 (Exact Endo–Duhamel reduction of the ghost determinant)**.** *With the positive Euclidean convention $H = -M_{\text{eff}}$, set*

$$F = -\Box\mathbf{1} + \mathrm{Ric}, \qquad H_0 = F - \frac{1}{2}\nabla\nabla\cdot, \qquad W = -2\,\mathrm{Ric}\,.$$

Then $H = H_0 + W$ exactly. The base H_0 is the nondegenerate Endo vector operator with parameter $\alpha = -1/2$, whose transverse and longitudinal principal eigenvalues are 1 and $3/2$. The Ward identities for F give the exact heat kernel [12]

$$K_{H_0}(t) = K_F(t) - \nabla\nabla' \int_t^{3t/2} ds\, K_{\Delta_0}(s).$$

Thus, on nonzero modes of a compact background without boundary,

$$\det{}' H_0 = \det{}' F\, \frac{\det{}'((3/2)\Delta_0)}{\det{}'\Delta_0},$$

and the last ratio contributes only the local zeta-scaling term $\zeta_{\Delta_0}(0)\log(3/2)$.

Since $W = O(\mathcal{R})$, the Duhamel expansion through $O(\mathcal{R}^3)$ terminates, at the declared order, after the one-, two- and three-W insertion traces. More precisely, the zero-, one-, two- and three-insertion rows require the Endo kernels only through curvature orders $3, 2, 1, 0$, respectively. This converts the missing generic ghost calculation into a finite coefficient problem. The insertion traces and their carrier coordinates are not yet complete. The computational supplement now reduces the generic nonexceptional-momentum three-insertion triangle to eight exact projector sectors and twenty rational simplex/Wick rows. On the declared scalar-flat inverse domain, the contracted Weyl and Bianchi identities independently fix

$$K_{\mu\nu} = \frac{2}{\Box}\nabla^\beta\nabla^\alpha C_{\alpha\mu\beta\nu} = R_{\mu\nu} + O(\mathcal{R}^2),$$

so the cubic replacement of three labelled K tensors by three labelled Ricci tensors is exact modulo $O(\mathcal{R}^4)$. A direct flat-TT tensor fixture fixes the sign. This closes the carrier normalization needed by the tensor projection: the zero-derivative sector can feed I_{10} and the longitudinal sectors can feed $I_{24}, I_{25}, I_{28}, I_{29}$. The next proposition performs that projection. The curved-Endo one-/two-insertion rows are reduced to minimal resolvent carriers below; their evaluation and the generic physical fourth-order Hessian kernel remain separate open calculations.

Proposition 6.11 (Exact Hodge–resolvent reduction of the ghost $n = 1, 2$ rows)**.** *On primed nonzero modes, integration of the exact Endo heat-kernel identity gives*

$$G_{H_0} = G_F - \frac{1}{3}L, \qquad L = d\Delta_0^{-2}\delta.$$

Indeed, after exchanging the proper-time integrations the domain is $2s/3 \le t \le s$, of width $s/3$, and $\int_0^\infty sK_{\Delta_0}(s)\,ds = \Delta_0^{-2}$. Writing $W = -2\,\mathrm{Ric}$, cyclicity therefore reduces the complete one- and two-insertion nonminimal architecture to the following five minimal vector/scalar carriers:

$$\begin{aligned}
\Gamma^{(1)}_{mgh} &= \mathrm{Tr}_1(G_F W) - \frac{1}{3}\,\mathrm{Tr}_0(\Delta_0^{-2}\delta W d),\\
\Gamma^{(2)}_{mgh} &= -\frac{1}{2}\,\mathrm{Tr}_1(G_F W G_F W) + \frac{1}{3}\,\mathrm{Tr}_0(\Delta_0^{-2}\delta W G_F W d)\\
&\qquad - \frac{1}{18}\,\mathrm{Tr}_0(\Delta_0^{-2}\delta W d\,\Delta_0^{-2}\delta W d).
\end{aligned}$$

The coefficients $(1, -1/3)$ and $(-1/2, 1/3, -1/18)$ are exact. An independent rational fixture uses a symmetric, noncommuting W that mixes the transverse and longitudinal sectors and reproduces the direct resolvent traces through $n = 3$.

This closes the nonminimal reduction. The next proposition evaluates the combined pure-vector slice; the three carriers containing $\delta W d$ remain outside the imported minimal-potential kernels.

Proposition 6.12 (Pure-vector $n = 1+n = 2$ CPT projection and minimal missing-carrier theorem)**.** *On the scalar-flat source complement, the positive minimal vector operators $F = -\Box + \mathrm{Ric}$ and $F + W = -\Box - \mathrm{Ric}$ correspond in the CPT-IV convention to $P_F = -\,\mathrm{Ric}$ and $P_H = +\,\mathrm{Ric}$. Even-P rows cancel and single-P scalar traces vanish. The combined physical pure-vector contribution is therefore exactly*

$$6\Gamma_1\mathcal{S}_1 - 2\Gamma_3\mathcal{S}_3 - 2\Gamma_{14}\mathcal{S}_{14},$$

where $\Gamma_1, \Gamma_3, \Gamma_{14}$ are the hash-pinned CPT-IV rows 1, 3 and 14. In the ordered channel basis

$$(I_{10}; I_{24}^{123}, I_{24}^{213}, I_{24}^{312}; I_{25}^{123}, I_{25}^{213}, I_{25}^{312}; I_{28}^{123}, I_{28}^{132}, I_{28}^{231}; I_{29}),$$

the two nontrivial source structures project as

$$\begin{aligned}
\mathcal{S}_3 &\longmapsto \left(-2, -\frac{1}{x_2}, -\frac{1}{x_1}, -\frac{x_1 + x_2}{x_1 x_2}, -\frac{2}{x_1}, -\frac{2}{x_2}, 0, 0, 0, 0, 0\right),\\
\mathcal{S}_{14} &\longmapsto \left(-\frac{x_1 + x_2 - x_3}{2}, 0, 0, -1, -1, -1, 0, 0, 0, 0, 0\right).
\end{aligned}$$

Ten exact momentum fixtures and all 5^3 transverse-tracefree products verify 2,500 projection identities; 750 further rows fix the Riemann/Ricci and contracted-Bianchi convention. An independent unseen-fixture replay rejects a coordinate mutation.

The remaining three Hodge carriers contain

$$D_W = \delta W d, \qquad \sigma_2(D_W)(p) = W^{\mu\nu}p_\mu p_\nu.$$

This is a curvature-dependent anisotropic principal-symbol insertion, not a minimal bundle potential P; the mixed carrier also couples scalar and vector resolvents. Hence the imported minimal-Laplace CPT rows do not evaluate these three carriers separately. The next proposition resums them into one scalar Schur determinant; its generic kernel, the physical fourth-order Hessian, and the complete repository functions remain open.

Proposition 6.13 (Exact longitudinal ghost Schur resummation)**.** *On the same primed nonzero-mode domain, write*

$$A = F + W, \qquad H = A + \frac{1}{2}d\delta, \qquad H_0 = F + \frac{1}{2}d\delta.$$

The Ward identities $Fd = d\Delta_0$, $\delta F = \Delta_0\delta$, and $\delta d = \Delta_0$ give the normalized scalar Schur operator

$$S_L(W) = \frac{2}{3}\mathbf{1} + \frac{1}{3}\delta(F+W)^{-1}d, \qquad \sigma_0(S_L) = 1.$$

For one compatible determinant-class relative prescription, the matrix determinant lemma gives

$$\mathrm{Det}_{\mathrm{rel}}(H, H_0) = \mathrm{Det}_{\mathrm{rel}}(F+W, F)\,\mathrm{Det}_F(S_L(W), \mathbf{1}).$$

If

$$B_n = \Delta_0^{-1}\delta W(G_F W)^{n-1} d\Delta_0^{-1},$$

then $S_L = \mathbf{1} + \sum_{n\geq 1}(-1)^n B_n/3$, and its logarithm begins

$$\begin{aligned} O(W) &: -\frac{1}{3}\operatorname{Tr} B_1, \\ O(W^2) &: \frac{1}{3}\operatorname{Tr} B_2 - \frac{1}{18}\operatorname{Tr} B_1^2, \\ O(W^3) &: -\frac{1}{3}\operatorname{Tr} B_3 + \frac{1}{9}\operatorname{Tr}(B_1 B_2) - \frac{1}{81}\operatorname{Tr} B_1^3. \end{aligned}$$

The first two lines reproduce the three previously isolated longitudinal carriers, and the third fixes their complete cubic weights. A distinct noncommuting exact fixture verifies the determinant and trace-log identities; on an Einstein background the minimal-vector longitudinal factor times S_L reproduces $(\Delta_0 - R/3)/\Delta_0$.

This is a resummation theorem, not a form-factor evaluation. A separately zeta-regularized factorization can carry a local multiplicative anomaly, which is not evaluated here. Moreover $S_L - \mathbf{1}$ starts at pseudodifferential order -2 in four dimensions, so ordinary trace-class Fredholm determinacy is not implied by order counting. A regularized relative determinant or equivalent common trace regulator, together with the possible local term, is the remaining longitudinal coefficient input.

Proposition 6.14 (Sharp Schatten split and critical Schur residue)**.** *Set $K = S_L - \mathbf{1}$ on a closed compact four-manifold, with the primed zero-mode complement and elliptic inverses fixed. The exact resolvent identity gives*

$$K = -\frac{1}{3}\delta(F+W)^{-1}W d\Delta_0^{-1}, \qquad \sigma_{-2}(K)(x,\xi) = -\frac{1}{3}\frac{\langle \xi, W\xi\rangle}{|\xi|^4}.$$

Hence K is a classical scalar pseudodifferential operator of order -2. Its singular values obey $s_j(K) = O(j^{-1/2})$ [16], so $K \in \mathcal{S}_p$ for every $p > 2$, in particular $K \in \mathcal{S}_3$; K^3 is trace class. The third modified Fredholm determinant [15]

$$\det{}_3(\mathbf{1}+K) = \det\left[(\mathbf{1}+K)e^{-K+K^2/2}\right]$$

is therefore canonical. For a declared local trace extension $\mathcal{R}$, the remaining renormalized split is

$$\log \mathrm{Det}_{3,\mathcal{R}}(\mathbf{1}+K) = \mathcal{R}(K) - \frac{1}{2}\mathcal{R}(K^2) + \log\det{}_3(\mathbf{1}+K).$$

The critical order--4 residues are already fixed. In the convention

$$\mathrm{Wres}\, P = (2\pi)^{-4}\int_M \int_{|\xi|=1} \sigma_{-4}(P)\,\mathrm{d}S\,\mathrm{dvol},$$

the four-sphere moment gives

$$\operatorname{Wres}(K^2) = \frac{1}{(4\pi)^2}\int_M \frac{(\operatorname{tr} W)^2 + 2\operatorname{tr}(W^2)}{108} = \frac{1}{(4\pi)^2}\int_M \frac{R^2 + 2R_{\mu\nu}R^{\mu\nu}}{27}.$$

The residue of K itself needs one subprincipal row. Through residue order, the resolvent expansion is

$$K = -\frac{1}{3}B_1 + \frac{1}{3}B_2 + O(\Psi^{-6}), \qquad B_1 = \Delta_0^{-1}\delta W d\Delta_0^{-1}, \quad B_2 = \Delta_0^{-1}\delta W G_F W d\Delta_0^{-1}.$$

Cyclicity and the mixed scalar heat kernel, with the standard heat/residue normalization [17, 18], give

$$\operatorname{Wres}(B_1) = \frac{1}{(4\pi)^2}\int_M \left[-\frac{1}{3}W^{\mu\nu}R_{\mu\nu} + \frac{1}{6}(\operatorname{tr} W)R\right], \qquad \operatorname{Wres}(B_2) = \frac{1}{(4\pi)^2}\int_M \frac{1}{2}\operatorname{tr}(W^2).$$

Thus, for the repository insertion $W = -2\,\mathrm{Ric}$,

$$\operatorname{Wres}(K) = \frac{1}{(4\pi)^2}\int_M \frac{R^2 + 4R_{\mu\nu}R^{\mu\nu}}{9},$$

and the logarithmic residue is

$$\operatorname{Wres}\log S_L = \operatorname{Wres}(K) - \frac{1}{2}\operatorname{Wres}(K^2) = \frac{1}{(4\pi)^2}\int_M \frac{5R^2 + 22R_{\mu\nu}R^{\mu\nu}}{54}.$$

On an Einstein background the exact ratio $S_L = (\Delta_0 - R/3)(\Delta_0 - R/2)^{-1}$ independently gives $\operatorname{Wres}(K) = (4\pi)^{-2}\int 2R^2/9$*. A covariantly constant isotropic test $W = w\mathbf{1}$ likewise reduces B_1 to $w\Delta_0^{-1}$. These checks fix the mixed-kernel sign and normalization.*

These equations compute the canonical local residues, not the full Schur determinant. The regulated finite value $\mathcal{R}(K)$, the finite part of $\mathcal{R}(K^2)$, and any zeta multiplicative anomaly remain open. Converting $\operatorname{Wres}\log S_L$ *into a pole or scale coefficient requires the order and normalization of a chosen reference operator; the next proposition makes that choice explicitly.*

Proposition 6.15 (Order-two weighted-trace pole and scale row)**.** *On the primed scalar space let*

$$Q = \Delta_0 + \Pi_0, \qquad \operatorname{ord} Q = 2, \qquad Q_\mu = Q/\mu^2,$$

where the smoothing projector only restores invertibility. Declare

$$\mathcal{R}_\mu(A) = \operatorname*{FP}_{z=0}\, \mathrm{TR}'\big[AQ_\mu^{-z}\big].$$

The weighted-trace Laurent formula [19] gives

$$\mathrm{TR}'(AQ^{-z}) = \frac{\operatorname{Wres}(A)}{2z} + \mathcal{R}_Q(A) + O(z).$$

Since $Q_\mu^{-z} = \mu^{2z}Q^{-z}$, multiplication of the Laurent series fixes both normalizations:

$$\mathcal{R}_{e^t\mu}(A) - \mathcal{R}_\mu(A) = t\operatorname{Wres}(A).$$

In particular,

$$\operatorname*{Res}_{z=0}\, \mathrm{TR}'\big[(\log S_L)Q^{-z}\big] = \frac{1}{(4\pi)^2}\int_M \frac{5R^2 + 22R_{\mu\nu}R^{\mu\nu}}{108},$$

and

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}\log\mu}\log\mathrm{Det}_{3,\mathcal{R}_\mu}(S_L) = \mathrm{Wres}\log S_L = \frac{1}{(4\pi)^2}\int_M \frac{5R^2+22R_{\mu\nu}R^{\mu\nu}}{54}.}$$

This closes the reference-specific pole and scale conversion. It does not fix the reference-scale constants $\mathcal{R}_{\mu_0}(K)$ and $\mathrm{FP}\,\mathcal{R}_{\mu_0}(K^2)$, nor the local multiplicative anomaly of a separately factorized zeta prescription [20].

Proposition 6.16 (Exact round-S^4 finite Schur benchmark)**.** *On the round unit four-sphere, delete the absent constant-gradient row and the five $\ell = 1$ proper-conformal-Killing ghost zero modes. On the remaining $\ell \geq 2$ carrier,*

$$\lambda_\ell = \ell(\ell+3), \qquad d_\ell = \frac{(2\ell+3)(\ell+2)(\ell+1)}{6}, \qquad K_\ell = \frac{2}{\lambda_\ell - 6}.$$

At $\mu_0 = 1$ in inverse-radius units, set

$$u_\pm = \frac{7\pm\sqrt{33}}{2}, qquad \Psi = \psi(u_-) + \psi(u_+), \qquad \Psi_1 = \psi^{(1)}(u_-) - \psi^{(1)}(u_+).$$

Hurwitz-zeta continuation of the complete spectrum [21] and the same-order weight-change formula give

$$\boxed{\mathcal{R}_{\Delta_0}(K) = -\frac{20}{9} - \frac{8}{3}\Psi,}$$

$$\boxed{\mathrm{FP}\,\mathcal{R}_{\Delta_0}(K^2) = -\frac{2}{3}\Psi + \frac{16}{3\sqrt{33}}\Psi_1.}$$

Numerically these are $-3.0967576144286354\ldots$ and $2.7591028732128106\ldots$. Hence the finite low-order part of the modified-determinant split is

$$\mathcal{R}_{\Delta_0}(K) - \frac{1}{2}\,\mathrm{FP}\,\mathcal{R}_{\Delta_0}(K^2) = -4.4763090510350407\ldots.$$

The remaining canonical tail is absolutely convergent. Exact alternating- series bounds on the first nineteen modes, followed by a rational Euler–Maclaurin enclosure of the Hurwitz-zeta tail, give

$$\boxed{\log\det{}_3(I+K) = 0.49816356541962909843125329994148187238611929 34\ldots,}$$

with certified enclosure width below 5.8×10^{-48}. Hence, for this declared weight and reference scale,

$$\boxed{\log\mathrm{Det}_{3,\mathcal{R}_{\Delta_0}}(S_L) = -3.9781454856154116274753955548059869205821661933\ldots.}$$

This is a special-background benchmark, not the generic finite row. Indeed, $K \mapsto K+T$ for arbitrary finite-rank smoothing T preserves every homogeneous symbol and Wodzicki residue but shifts the two finite rows by $\mathrm{Tr}\,T$ and $\mathrm{Tr}(KT+TK+T^2)$. Thus the generic values require the full primed Green kernel or equivalent spectral measure. Any separately factorized multiplicative anomaly remains open.

Proposition 6.17 (Exact ghost $n = 3$ scalar-flat five-carrier projection)**.** *Evaluate the eleven effective labelled orientations*

$$I_{10}; \quad I_{24}^{\times 3}; \quad I_{25}^{\times 3}; \quad I_{28}^{\times 3}; \quad I_{29}$$

on the $5^3 = 125$ *products of exact transverse-tracefree polarizations at generic nonexceptional Euclidean momenta. The resulting* 125×11 *matrix has exact rank ten. Appending the quotient section*

$$\Gamma_{28}^{123} + \Gamma_{28}^{132} + \Gamma_{28}^{231} = 0$$

fixes the unique CPT-IV four-dimensional null direction. In that section, the complete three-insertion ghost triangle has exact channel coordinates

$$\Gamma_i(\alpha; x) = \frac{N_i(\alpha_1, \alpha_2; x_1, x_2, x_3)}{\Delta^4}, \qquad \Delta = \alpha_0\alpha_1 x_1 + \alpha_1\alpha_2 x_2 + \alpha_2\alpha_0 x_3,$$

where every coefficient of every numerator N_i *is rational. If carrier* i *has* d_i *explicit derivatives,* N_i *is homogeneous of box degree* $3 - d_i/2$ *and has alpha degree at most nine. The* $W = -2\,\mathrm{Ric}$ *and third-order trace-log multiplier* $-8/3$ *is included; the common* $(4\pi)^{-2}$ *loop prefactor is not.*

Ten box-unisolvent momentum fixtures reconstruct the stored polynomials. Two unseen momentum fixtures and unseen rational simplex points independently reproduce all 125 triangle amplitudes with zero exact residual, and a rehashed coefficient mutation is rejected. This is the parametric $n = 3$ *ghost contribution. It is not the integrated five repository functions, the evaluated five minimal* $n = 1/n = 2$ *resolvent carriers, the complete ghost determinant, or the physical fourth-order Hessian contribution.*

Proposition 6.18 (Exact local $\Box R$ scheme conversion)**.** *In the zeta/proper-time prescription of Barvinsky et al. [7], the imported fourth heat-kernel rows combine with weights* $(2, -1, -2)$ *to give*

$$b_{\Box}^{\mathrm{raw}} = \frac{7}{2}\log\frac{3}{2} - \frac{159}{80} < 0.$$

The project convention

$$\omega\Box R = -\frac{1}{12}s(R^2) \pmod{\mathrm{d}}$$

therefore fixes the one-loop strict-metric counterterm

$$z_R^{\mathrm{raw}\to 0} = \frac{b_{\Box}^{\mathrm{raw}}}{12} = \frac{7}{24}\log\frac{3}{2} - \frac{53}{320}.$$

It converts the raw coefficient to the repository $\Box R = 0$ *convention and, when added to the raw anomaly-induced local* R^2 *coefficient* $391/960 - (7/24)\log(3/2)$*, gives exactly* $29/120$*. This is a relative local one-loop scheme conversion. By itself it neither proves the preceding FV Ricci-sector dependence theorem nor fixes the independent dressed* $R(\widehat{g})^2$ *counterterm in the compensator theory, and it is not an all-loop equivalence theorem for strict Weyl gravity.*

7 What does not follow

The strict obstruction and extended restoration are local one-loop results. They do not establish:

- an all-loop extended QME;
- a Lorentzian renormalized time-ordered product or Slavnov identity;
- a BRST-compatible global Hadamard state;
- positivity, a particle Hilbert space, scattering, or unitarity;

- survival or removal of either residual deformation class;
- a quantum Cartan identity or residual moment-map algebra.

Residual transfer is specifically forbidden at present. The frozen classical contraction belongs to the strict theory and has no compensator rows. The local Wess–Zumino contribution to Q_1 is fixed, but the proposition proves that it is not a unique complete renormalized operator. The anomaly-induced Paneitz/Riegert representative fixes one nonlocal solution, and the logarithmic conformal-sector coefficient is covariant and universal through curvature order two. The selected logarithmic carrier now also has the exact Fradkin–Vilkovisky Weyl-orbit completion above. The relative strict-metric raw-to-$\Box R = 0$ R^2 scheme shift is now exact. The FV anomaly action and structural dependence of its Ricci-scalar sector are also exact. The additive finite C^2 constant and absolute dressed $R(\widehat{g})^2$ normalization remain open. The zero-derivative algebraic cubic basis, the parity-even third-curvature five-carrier quotient and the five exact universal CPT source kernels are exact. The generic ghost calculation now proves that direct minimal-CPT substitution is obstructed by a nonminimal, Hodge-mixed Schur operator. The exact Endo reduction leaves finite one-, two- and three-Ricci insertion traces. The three-insertion row is now reduced at generic nonexceptional momentum to an exact eight-sector Feynman-simplex/Wick kernel and projected onto the ten-dimensional scalar-flat five-carrier quotient as exact rational simplex integrands. The one-/two-insertion nonminimal architecture is now reduced exactly to five minimal vector/scalar resolvent carriers. Their pure-vector physical sum is evaluated exactly from CPT rows 1, 3 and 14. The three longitudinal D_W towers are exactly resummed into one normalized scalar Schur determinant. Its order--2 correction is proved to lie in $\mathcal{S}_3$, and $\mathrm{Wres}(K)$, $\mathrm{Wres}(K^2)$, and $\mathrm{Wres} \log S_L$ are exact. The round-S^4 reference finite rows, canonical $\det_3$ tail, and selected weighted modified determinant are complete as well, but the smoothing theorem shows that their generic counterparts require a full primed Green/spectral carrier. That carrier, the possible local zeta term, and physical kernel, hence the five complete repository form-factor functions and coefficients, the parity-odd derivative manifest, global Green/kernel data, and renormalized-product contribution remain open. The next gate is a compensator-inclusive classical contraction together with those missing data. The exact positive-Berger contraction does not fill this gate: its row named τ is a temporal diffeomorphism ghost on a different background, not the Wess–Zumino scalar compensator. The separate same-background free-state and Bridge 4 programme is outside this paper. Bridge 5 has a QME disposition but still lacks the independent interaction-to-physical-branch map. No classical tangent obstruction is therefore called ghost removal or a quantum constraint.

8 Machine certification and reproduction

The coefficient-bearing theorem receipts are

- `quantum-weyl/local_bv/certificates/AFNO_H14_EVEN_CANONICAL_QUOTIENT.json`.
- `quantum-weyl/local_bv/certificates/AFNO_H14_ODD_CANONICAL_QUOTIENT.json`.
- `quantum-weyl/local_bv/certificates/AFNO_DIFF_MIXED_MINIMAL_BV_H14.json`.
- `quantum-weyl/local_bv/certificates/MINIMAL_BV_KOSZUL_TATE_COLLAPSE.json`.
- `quantum-weyl/local_bv/certificates/GENERAL_NONMINIMAL_GAUGE_FIXED_CONTRACTION.json`.

- quantum-weyl/spectral/euclidean/certificates/REPOSITORY_EUCLIDEAN_ELLIPTIC_COMPLEX.json.
- quantum-weyl/spectral/euclidean/certificates/REPOSITORY_FULL_BV_MULTIPLICITY_LEDGER.json.
- quantum-weyl/spectral/euclidean/certificates/STANDARD_EUCLIDEAN_LOCAL_B4_INTEGRATION_SLICE.json.
- quantum-weyl/spectral/euclidean/certificates/REPOSITORY_NONCONFORMALLY_FLAT_OR_RICCI_FLAT_FULL_BV_OPERATOR_MEASURE_COEFFICIENT_MATCH.json.
- quantum-weyl/anomalies/certificates/REGULATED_REPOSITORY_BV_SLAVNOV_BREAKING.json.
- quantum-weyl/anomalies/certificates/UNITARY_CONFORMAL_MATTER_CANCELLATION_NO_GO.json.
- quantum-weyl/anomalies/certificates/WESS_ZUMINO_MINIMAL_BV_COTANGENT_LIFT.json.
- quantum-weyl/anomalies/certificates/WESS_ZUMINO_EXTENDED_LOCAL_BV_COHOMOLOGY.json.
- quantum-weyl/anomalies/certificates/WESS_ZUMINO_COMPENSATOR_EXTENSION_PREFLIGHT.json.
- quantum-weyl/transfer/certificates/ONE_LOOP_SLAVNOV_Q1_DISPOSITION.json.
- quantum-weyl/transfer/certificates/ANOMALY_INDUCED_NONLOCAL_GAMMA1.json.
- quantum-weyl/transfer/certificates/FLAT_TT_LOGARITHMIC_GAMMA1.json.
- quantum-weyl/transfer/certificates/CURVATURE_SQUARED_COVARIANT_LOG_GAMMA1.json.
- quantum-weyl/transfer/certificates/FV_CONFORMIZED_C2_LOG_GAMMA1.json.
- quantum-weyl/transfer/certificates/FV_ANOMALY_ACTION_RICCI_SECTOR.json.
- quantum-weyl/transfer/certificates/FOUR_DIMENSIONAL_ALGEBRAIC_CUBIC_WEYL_CARRIERS.json.
- quantum-weyl/transfer/certificates/FOUR_DIMENSIONAL_THIRD_CURVATURE_WEYL_CARRIER_MANIFEST.json.
- quantum-weyl/transfer/certificates/CPT_UNIVERSAL_THIRD_CURVATURE_KERNELS.json.
- quantum-weyl/spectral/euclidean/certificates/GENERIC_BACKGROUND_GHOST_N1_N2_VECTOR_CPT_PROJECTION.json.
- quantum-weyl/spectral/euclidean/certificates/GENERIC_BACKGROUND_DIFF_WEYL_GHOST_CPT_OBSTRUCTION.json.

- quantum-weyl/spectral/euclidean/certificates/GENERIC_BACKGROUND_GHOST_ENDO_DUHAMEL_REDUCTION.json.
- quantum-weyl/spectral/euclidean/certificates/GENERIC_BACKGROUND_GHOST_N1_N2_HODGE_RESOLVENT_REDUCTION.json.
- quantum-weyl/spectral/euclidean/certificates/GENERIC_BACKGROUND_GHOST_LONGITUDINAL_SCHUR_RESUMMATION.json.
- quantum-weyl/spectral/euclidean/certificates/GENERIC_BACKGROUND_GHOST_SCHUR_SCHATTEN_SPLIT.json.
- quantum-weyl/spectral/euclidean/certificates/GENERIC_BACKGROUND_GHOST_SCHUR_WODZICKI_RESIDUE.json.
- quantum-weyl/spectral/euclidean/certificates/GENERIC_BACKGROUND_GHOST_SCHUR_WEIGHTED_TRACE_SCALE.json.
- quantum-weyl/spectral/euclidean/certificates/ROUND_S4_GHOST_SCHUR_FINITE_WEIGHTED_TRACES.json.
- quantum-weyl/spectral/euclidean/certificates/GENERIC_BACKGROUND_GHOST_N3_ADIABATIC_CARRIER.json.
- quantum-weyl/spectral/euclidean/certificates/GENERIC_BACKGROUND_GHOST_N3_TRIANGLE_KERNEL.json.
- quantum-weyl/transfer/certificates/SCALAR_FLAT_K_RICCI_CUBIC_CROSSWALK.json.
- quantum-weyl/spectral/euclidean/certificates/GENERIC_BACKGROUND_GHOST_N3_FIVE_CARRIER_PROJECTION.json.
- quantum-weyl/spectral/euclidean/certificates/WEYL_GRAVITON_BOX_R_SCHEME_CONVERSION.json.

The companion computational supplement prints the exact quotient ranks, orbit counts, coefficient vector, quartet matrices, extended boundary matrix, and input-hash ledger from generated certificate tables. The claim map binds the manuscript, both PDFs, the generated tables, their producer and verifier, and all explicit nonclaims.

The scoped replay is:

```
PYTHONPATH=quantum-weyl python3 -m \
  local_bv.verify_diff_mixed_total_complex
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_standard_integration_slice
PYTHONPATH=quantum-weyl python3 -m \
  anomalies.verify_repository_regulated_slavnov_breaking
PYTHONPATH=quantum-weyl python3 -m \
  anomalies.verify_unitary_matter_cancellation_no_go
PYTHONPATH=quantum-weyl python3 -m \
  anomalies.verify_wess_zumino_minimal_bv_cotangent_lift
PYTHONPATH=quantum-weyl python3 -m \
  anomalies.verify_wess_zumino_extended_local_bv_cohomology
PYTHONPATH=quantum-weyl python3 -m \
```

```
  transfer.verify_one_loop_slavnov_q1_disposition
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_flat_tt_logarithmic_gamma1
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_curvature_squared_covariant_log_gamma1
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_fv_conformized_c2_log_gamma1
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_fv_anomaly_action_ricci_sector
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_cubic_weyl_carrier_basis
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_third_curvature_weyl_manifest
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_cpt_universal_third_curvature_kernels
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_generic_background_ghost_cpt_obstruction
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_generic_background_ghost_endo_duhamel_reduction
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_generic_background_ghost_n1_n2_hodge_resolvent_reduction
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_generic_background_ghost_n1_n2_vector_cpt_projection
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_generic_background_ghost_longitudinal_schur_resummation
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_generic_background_ghost_schur_schatten_split
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_generic_background_ghost_schur_wodzicki_residue
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_generic_background_ghost_schur_weighted_trace_scale
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_round_s4_ghost_schur_finite_weighted_traces
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_generic_background_ghost_n3_adiabatic_carrier
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_generic_background_ghost_n3_triangle_kernel
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_generic_background_ghost_n3_five_carrier_projection
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_scalar_flat_k_ricci_crosswalk
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_box_r_scheme_conversion
PYTHONPATH=quantum-weyl python3 -m verify_active_frontier
PYTHONPATH=quantum-weyl python3 -m atlas.verify_quantum_atlas_fragment
python3 paper/verify_12_pure_weyl_one_loop_bv_anomaly_claim_map.py
```

9 Outlook

The immediate local-to-residual question is now sharp. Supply the same-gauge generic-background full-BV Hessian and trace substitutions needed to match the five imported universal CPT kernels to repository parity-even third- curvature form-factor functions and coefficients on the certified carrier quotient, and classify the parity-odd derivative sector; then supply finite C^2 and absolute dressed $R(\widehat{g})^2$ normalization conditions and renormalized BV operator data to fix a complete Q_1; then, given an extended contraction $(\iota_{\rm ext}, \pi_{\rm ext}, S_{\rm ext})$, compute

$$q_1 = \pi_{\rm ext} Q_1 \iota_{\rm ext}$$

and test first-order nilpotency, the quantum Cartan defect, incoming $H^3 \to H^4$, outgoing $H^4 \to H^5$, operator mixing, and the corrected pairing. This is a new theorem, not a corollary of local QME restoration.

The independent Lorentzian rail must construct renormalized causal products and a compatible state before any unitary or particle conclusion is possible. The present result nevertheless fixes the finite local obstruction vector that every such construction must reproduce or trivialize.

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